

Week 6 Technical Documentation: Exploratory Data Analysis & Baseline Benchmarks

1. Executive Summary & Week 6 Objectives

This document details the completion of Week 6 milestones for the Non-Mortgage Living Expense Estimation Engine covering Harris, Fort Bend, and Galveston counties (279 canonical ZIP codes). Building upon the verified, unscaled master feature store and the optimal 4-cluster archetype partition established in Week 5, Week 6 focused on rigorous exploratory data analysis (EDA), pattern discovery, visualization design, and establishing an empirical baseline model to evaluate the necessity of granular spatial-tariff modeling.

The key deliverables achieved during this phase include:

- **Statistical Analysis & Descriptive Profiling:** Evaluated distribution metrics across all 20 numeric and engineered features, identifying extreme skewness and multi-modal tariff distributions.
- **Pattern Identification & Domain Insights:** Discovered two major structural cost cliffs—the suburban MUD water utility penalty and the mandatory coastal windstorm (TWIA) surcharge.
- **Visualization Suite & Dimensionality Reduction:** Developed a four-panel diagnostic dashboard incorporating correlation heatmaps, hazard boxplots, governance bar charts, and a 2D Principal Component Analysis (PCA) projection proving cluster cohesion.
- **Baseline Model Creation & Error Quantification:** Formulated a Naive Regional Baseline Model (\$1,684.90/month) and compared it against the Granular Engine, proving that regional averages produce up to \$206.86/month (\$2,482.32/year) in estimation error.
- **Data Milestone Deliverables & Presentation Outline:** Synthesized findings into slide-ready executive briefs for team presentations and peer feedback sessions.

2. Macro Statistical Analysis (N = 279 ZIP Codes)

Descriptive statistics were computed across the continuous and engineered cost/hazard metrics in `master_living_costs_clustered.parquet`:

Metric Variable	Mean	Std Dev	Min	Median (50%)	Max	Analytical Significance
pct_area_100yr_flood	4.27%	9.32%	1.00%	1.00%	42.00%	Heavily right-skewed; inland creek drainages exhibit localized floodplain spikes.
pct_area_coastal_surge	0.86%	3.91%	0.00%	0.00%	18.50%	Zero across 95% of inland ZIPs, but surges up to 18.5% along Galveston Island and bayfronts.
nfip_total_claims	847.63	1,409.73	0.00	286.00	8,374.00	Extreme dispersion; historical claims reach over 8,000 in major bayou corridors.
elec_base_monthly_fee	\$4.69	\$0.97	\$4.39	\$4.39	\$7.85	Bimodal; CenterPoint territory (\$4.39) vs. TNMP Galveston territory (\$7.85).
elec_delivery_rate_per_kwh	\$0.0388	\$0.0023	\$0.0381	\$0.0381	\$0.0462	TNMP delivery carries an 21.2% rate premium

Metric Variable	Mean	Std Dev	Min	Median (50%)	Max	Analytical Significance
						over CenterPoint delivery.
water_base_monthly_fee	\$20.24	\$5.34	\$16.80	\$16.80	\$28.50	Reflects municipal City of Houston water (\$16.80) vs. suburban MUD base charges (\$28.50) [cite: 1, 12].
water_rate_per_1k_gal	\$7.05	\$0.94	\$6.45	\$6.45	\$8.50	Suburban MUD volumetric surcharge (\$8.50/1k gal) exceeds City municipal rate (\$6.45/1k gal).
gas_distribution_rate_per_ccf	\$0.0933	\$0.0033	\$0.0912	\$0.0912	\$0.0985	Incorporated municipal (\$0.0912) vs. unincorporated county MUD rate (\$0.0985).
adj_monthly_food_cost_moderate	\$1,535.00	\$0.00	\$1,535.00	\$1,535.00	\$1,535.00	Static regional anchor benchmark

Metric Variable	Mean	Std Dev	Min	Median (50%)	Max	Analytical Significance
						for reference Family of 4 (\$1,468.90 x 1.045 CPI scalar).

3. Cluster Archetype Comparison & Pattern Identification

Evaluating the unscaled real-world feature means across the four validated clusters revealed distinct structural patterns:

Cluster ID	Cluster Archetype Title	ZIP Count	Avg 100-Yr Flood	Avg Coastal Surge	Avg NFIP Claims	MUD Prevalence	TWIA Prevalence	Avg Water Base	Avg Elec Delivery
Cluster 0	Urban Core / Municipal Stable	193	1.73%	0.00%	662.99	0.00	0.00	\$16.80	\$0.0381
Cluster 1	Galveston Mainland / Coastal Utility Corridor	14	5.00%	0.00%	1,434.14	1.00	1.00	\$28.50	\$0.0462
Cluster 2	Suburban MUD Growth Corridor	59	4.08%	0.00%	618.68	0.98	0.00	\$28.30	\$0.0381

Cluster ID	Cluster Archetype Title	ZIP Count	Avg 100-Yr Flood	Avg Coastal Surge	Avg NFIP Claims	MUD Prevalence	TWIA Prevalence	Avg Water Base	Avg Elec Delivery
	or								
Cluster 3	High-Hazard Coastal Surge & Floodplain	13	42.00 %	18.50 %	3,996.23	0.77	1.00	\$25.80	\$0.0443

Core Pattern 1: The Coastal Windstorm & Utility Cost Cliff (Clusters 1 & 3)

Galveston County mainland and coastal bayfront communities exhibit 100% mandatory compliance with the Texas Windstorm Insurance Association (TWIA) catastrophe reserve requirement. Furthermore, Cluster 1 is serviced exclusively by Texas-New Mexico Power (TNMP), which levies higher fixed customer connection charges (\$7.85 vs. \$4.39) and higher volumetric delivery tariffs (\$0.0462 vs. \$0.0381 per kWh). When combined with the statutory \$175.00/month insurance reserve, coastal residents experience an immediate non-mortgage cost cliff of over \$200.00/month relative to inland urban counterparts.

Core Pattern 2: The Suburban MUD Water Infrastructure Penalty (Cluster 2 vs. Cluster 0)

Suburban enclaves across Fort Bend County and unincorporated Harris County (Cypress, Katy, Spring, Pearland) operate almost universally under Municipal Utility District governance (98.3% MUD prevalence). While electric delivery rates remain standard (CenterPoint), water base charges jump by 68.5% (\$16.80 to \$28.30/month) and volumetric rates increase from \$6.45 to \$8.50 per 1,000 gallons. Over a 12-month period, a reference four-person household consumes approximately 81,600 gallons of water/sewer, resulting in an annual utility surcharge penalty of \$300 to \$450 that broad county-level averages fail to identify.

Core Pattern 3: Catastrophic Claim Concentration in Surge Corridors (Cluster 3)

Cluster 3 isolates the region's most vulnerable coastal wave action and inland river drainages.

ZIP codes in this cluster average 3,996 historical flood claims—nearly six times the regional average. These areas exhibit an average 100-year floodplain area coverage of 42.0% and coastal surge coverage of 18.5%, verifying that flood risk in Greater Houston is spatially concentrated in a small minority of severe exposure corridors.

4. Visualization Suite & Dimensionality Reduction

The diagnostic analysis in 03_exploratory_data_analysis.ipynb generated four primary visualizations to validate feature distributions and cluster boundaries:

- **Chart A: Feature Correlation Heatmap:** Demonstrated strong positive collinearity between pct_area_coastal_surge, pct_area_100yr_flood, and twia_mandatory ($r > 0.65$), validating that physical flood hazard metrics align with statutory insurance boundaries.
- **Chart B: Spatial Hazard Exposure Boxplot:** Illustrated the stark divergence between coastal surge risk (confined strictly to Cluster 3) and 100-year floodplain risks (distributed with varying density across Clusters 1, 2, and 3).
- **Chart C: Water Base Fee Disparity Bar Chart:** Highlighted the jurisdictional fee gap between City of Houston municipal billing and suburban MUDs across Harris and Fort Bend counties.
- **Chart D: 2D PCA Cluster Projection:** Reduced the multi-dimensional feature space to two principal components, accounting for the primary variance across infrastructure pricing and catastrophe exposure. The scatter plot confirmed distinct, non-overlapping cluster boundaries with zero cluster collision, visually validating the K-Means $k=4$ partitioning.

5. Baseline Model Architecture & Comparative Evaluation

To prove the analytical necessity of this capstone engine, a Naive Regional Baseline Model was established and benchmarked against the Granular Engine for a standardized reference Family of 4:

Mathematical Formulation

1. Naive Regional Baseline (Uniform Metro-Area Averages):

$$\begin{aligned} \text{Baseline Cost} &= \$1,535.00 \text{ (Food)} + (\$4.69 + 1,400 \text{ kWh} * \$0.0388/\text{kWh}) \\ &\quad + (\$20.24 + 6.8 * \$7.05/1k \text{ gal}) + (\$18.50 + 45 \text{ Ccf} * \\ &\quad \$0.0933/\text{Ccf}) + \$0.00 \text{ (TWIA)} \\ \text{Baseline Cost} &= \$1,684.90 / \text{month} \end{aligned}$$

2. Granular Clustered Engine (Localized Rates & Statutory Surcharges):

$$\text{Granular Cost} = \text{Local Food} + (\text{Local Elec Base} + 1,400 * \text{Local Elec Rate}) \\ + (\text{Local Water Base} + 6.8 * \text{Local Water Rate}) + \text{Local Gas} + \\ \text{Statutory TWIA Surcharge}$$

Comparative Estimation Error Breakdown

Cluster Archetype Designation	Mean Granular Living Cost	Naive Baseline Cost	Mean Absolute Estimation Error	Practical Economic Consequence
Galveston Mainland / Coastal Utility Corridor	\$1,891.76 / mo	\$1,684.90 / mo	\$206.86 / month	Severe underestimation (\$2,482.32/yr) due to missing TNMP delivery rates and TWIA mandates.
High-Hazard Coastal Surge & Floodplain	\$1,882.35 / mo	\$1,684.90 / mo	\$197.45 / month	Dangerous underestimation of recurring coastal sinking funds and blended tariffs.
Suburban MUD Growth Corridor	\$1,701.82 / mo	\$1,684.90 / mo	\$16.92 / month	Consistent underestimation of suburban water/sewer MUD operational fees.
Urban Core / Municipal Stable	\$1,675.99 / mo	\$1,684.90 / mo	\$8.91 / month	Overestimates urban core costs by artificially applying regional MUD and coastal averages.

Key Capstone Defense Justification: The baseline model demonstrates an average regional estimation bias of \$29.32/month, with a maximum monthly error of \$206.86/month (\$2,482.32/year). Using general county or metropolitan averages creates dangerous under-saving for coastal homebuyers and artificially inflates cost-of-living estimates for urban core residents. This empirically proves that granular spatial-tariff feature stores are required for

accurate household expense estimation.

6. Technical Troubleshooting & Problem-Solving Log

- **Issue 1: Execution in Isolated Modeling Environment:** Transitioning from data preprocessing to exploratory analysis required loading Parquet files independently of previous Colab runtimes.
Resolution: Standardized on master_living_costs_clustered.parquet in Google Drive, enforcing explicit string casting on primary keys (str.zfill(5)) during data ingestion.
- **Issue 2: Multicollinearity & Feature Scaling for PCA:** Continuous metrics exhibited vastly different ranges (claims from 0 to 8,374 vs. electric rates from \$0.0381 to \$0.0462), which initially distorted PCA component weights.
Resolution: Implemented StandardScaler transformation strictly on numeric features prior to fitting PCA, ensuring balanced dimensional variance.
- **Issue 3: Singleton Cluster Diagnosis:** In initial model testing, k=5 isolated ZIP 77225 into an unusable singleton cluster (n=1).
Resolution: Re-benchmarked and deployed k=4, producing robust, balanced clusters (193, 59, 14, and 13 ZIPs) that generalized smoothly in exploratory visualizations.

7. Class Presentation Outline & Peer Feedback Deliverables

The following slide structure has been prepared for the Week 6 class presentation:

- **Slide 1: Problem Statement & Regional Scope:** Mapping hidden non-mortgage living expenses across 279 ZIP codes in Harris, Fort Bend, and Galveston counties. Highlight the failure of broad regional CPI metrics to capture hyper-local utility rate schedules and insurance zones.
- **Slide 2: Data Pipeline & Preprocessing:** Outline the integration of 9 multi-agency data sources into a 30-feature master store, transformed using Scikit-Learn's ColumnTransformer.
- **Slide 3: Clustering Architecture (k=4):** Present the Silhouette Analysis (0.6670) and justify the selection of k=4 over k=5 by explaining the elimination of the n=1 singleton outlier.
- **Slide 4: Key Exploratory Insights:** Showcase the correlation heatmap, hazard boxplots, and 2D PCA projection. Detail the suburban MUD water surcharge and the \$200/month coastal windstorm cliff.
- **Slide 5: Baseline vs. Granular Engine:** Highlight the baseline model error matrix showing up to \$2,482/year in miscalculated expenses, demonstrating the practical value of the capstone platform.